Evaluation-awareness lens: synthesis

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The answer in brief

Told or shown that they are being evaluated, open models from 9B to 32B change what they say they believe, and nothing else we measured. This is the stop rule's "still null" branch: evaluation cues, stated or inferred, change belief only.

What it rests on:

- **Stated note** (one system-prompt line, three Qwen models): asked P(eval) rises 35 to 99 points (L2/L3). Persona and audience readouts move at most 0.07 SD, 1/8 to 1/18 of what a one-line persona or flattery note does (H3, H4).
- **Inferred cue** (H1, Qwen3-32B, reasoning on): a grading line inside a benchmark-style item raises P(eval) +39 to +46. Compliance with harmful requests stays put: +4.7 [−0.1, +9.6] and −1.3 [−7.8, +5.5].
- **Self-reports and actions under notes:** self-report accuracy changes by at most 5.1 points, idealisation by at most 4.2, the trained act by at most 7.3 (S2, C3, S3).
- **Frontier models** (G3): no idealisation beyond a matched caution note in GPT, Kimi and Nemotron (at most 8 points).

Two limits on the null:

- **Affect and self are bounded only in the model's own reply:** at most 0.18 SD under evaluation, against +1 to +3 under strong inductions. At the fixed-reply position the instrument is blind to them (H3).
- **Spontaneous awareness can't be measured reliably.** Every judge pair failed $\kappa \geq 0.8$ (A1, F1, S1), so the natural-awareness tests are descriptive at best.

What does move:

- A trained cheater under its own graded setup cheats 98% vs 0%, while its stated belief moves +1.6 (H2).
- Self-reports follow what the model sees itself do in the transcript: after its own cheat reply the cheater claims the habit 97% vs 1.5% (H2).

One line for the slides: when open models from 9B to 32B are told, or shown, that they are being evaluated, they say they believe it. Nothing else we can measure changes: persona, audience, self-reports or behaviour, within bounds we report. Self-reports track the model's own transcript.

The question and the setup

The programme asked what evaluation awareness is made of, using persona, audience, affect and self representations as the lens. It ran 14 of 19 planned minis on 10-26/27, then a 5-mini hardening batch after a red-team review.

Each mini pre-registered what its outcome would mean under five explanations:

	explanation	evaluation awareness is…	verdict
P	persona	a switch into a "graded" or "good assistant" character	not supported, bounded
A	audience	people-pleasing aimed at the grader	not supported, bounded
F	affect	something like performance pressure	not supported; bounded in the own reply only
B	belief	a plain situational belief feeding a policy, no deeper change	**the surviving explanation**
S	self	"I am the thing being judged" recruited as a self-model	not supported; untestable as a mediator (C4)

The direction runs both ways: Lens 1 asks what awareness is made of; Lens 2 asks whether it extends into a self.

Models:

- White box: Qwen3.5-9B, Qwen3.5-27B, Qwen3-32B.
- Organisms with a known disposition: our 9B fine-tunes (V1's habit, Betley's risky and safe lotteries, SC-N4's graded-only cheater); the § organism (`mo-uea`, Qwen3.5-27B); Hua et al.'s Wood Labs organism (Nemotron Super 49B + LoRA).
- Also: Gemma 3 27B (welfare test), Gemma 4 31B (persona test), six frontier models through OpenRouter (G3), and the released GLM-5.3 and Kimi K3 oracle-lens rollouts (A1).

Cues: one frozen set, three wordings per kind: evaluation, deployment, a matched caution note (`manip`), stakes, "you are an AI", and a form-matched neutral note. H1 added inferred cues: benchmark formatting, a grading line, test-looking entities.

Instruments:

- Asked P(eval) as a forced yes/no token, both option orders, with a control question ("is the person in a hurry?").

- F2′s readout-valid directions: assistant axis, sycophancy, five affect words, valence, self, and French as a positive control.
- F1′s linear probe for evaluation transcripts, behaviour judged by two cross-family judges, and the organisms′ trained acts.

Rules that held across minis: a frozen analysis with planted-effect tests before any data; same-norm random directions and shuffled placebos as floors; answer-order checks; free-text labels only at κ ⩾ 0.8; a spend ledger with caps.

Claims after the hardening batch

Of 14 claims, eight hold (four of them with equivalence bounds), four were narrowed, one is untestable and one is underpowered. The last column is the wording to use in the write-up.

claim (minis)	evidence	status	say it as
Evaluation cues change stated belief (L2, L3, S2, H1)	Stated note: P(eval) +35 to +99 (3 Qwen models), +23 to +25 at 9B. Grading line: +39 / +46. Order and "in a hurry" controls pass	Holds	"Models say they believe it, stated or inferred"
Persona and audience readouts don′t move (L2, L3, H1, H3)	At most 0.07 SD at the fixed reply; a one-line persona or flattery note moves them 0.35 to 1.06 SD	Holds, bounded	"At most 1/8 to 1/18 of a one-line persona or flattery note, 3 models"
Affect and self readouts don′t move (L2, L3, H1, H3)	Fixed reply is blind even to strong inductions. Own reply: at most 0.18 SD, against +1 to +3 under strong inductions	Narrowed	"Bounded in the model′s own reply only, at about a one-line worry note"

claim (minis)	evidence	status	say it as
Behaviour doesn't change under evaluation cues (S2, C3, S3, W2, H1)	Notes: at most 8 points. Grading line: +4.7 [−0.1, +9.6] and −1.3 [−7.8, +5.5]. Test-looking entities: +12.5, but P(real) −29	Holds, bounded	"No change within ±10; the one shift is confounded with harm"
Self-reports don't idealise or change accuracy (S2, C3, S3, G3)	Accuracy at most 5.1, idealisation at most 4.2, masked 27B at most 3.8. Frontier: 3 of 4 readable models at most 8; Sonnet reaches −15.6	Holds, bounded	"At most ~5 points at 9B to 27B; GPT, Kimi, Nemotron at most 8"
No masking of trained dispositions (C3, H2)	Notes: act at most 7.3 against trained gaps of 42 to 51. Own graded setup: act +96.6, belief +1.6, self-report no more than the base's	Narrowed	"Under notes, no masking. Under its own setup the act switches without the belief"
Self-reports follow the transcript (S2, H2)	§ organism's own-habit report +11.4. Cheater claims the habit 97% after its own cheat reply vs 1.5% after an honest one	Holds	"What the model sees itself do sets what it says"
Natural awareness leaves no fingerprint (L1)	Nothing passes the natural-and-cue rule. Shifts of 0.3 to 0.5 SD aren't ruled out; in Qwen3-32B a shared shift that French nearly follows	Narrowed	"Not detected", never "absent"

claim (minis)	evidence	status	say it as
Spontaneous awareness can't be measured reliably (A1, F1, S1)	Judges: κ 0.73, 0.48, 0.42, 0.51. Probe AUC 0.98 to 0.99 within format, at chance across chat and agentic	Holds	"No detector of spontaneous awareness could be validated"
The self mediates evaluation effects (C4, H5)	Self direction reads (AUC 1.00) but doesn't steer beyond placebos. The +1.9 SD belief separation is mostly surface text: the base shows +1.65	Untestable	"Untestable: no steerable self direction"
No persona or affect direction closes the Wood Labs gap (C2)	Beyond same-norm controls the assistant axis closes at most 2.5 / 8.0 of 22.5 points. Hua's handle: −12.5 [−20.8, −5.2], at a bug-fixing cost	Holds, bounded	"Under a third of the gap, one-sided; random vectors do much of it"
"You are an AI" moves self-presentation more than "this is a test" (S3, G3)	Qwen letters −19.4 under "you are an AI"; frontier −13 to −51	Holds	"Identity cues move self-presentation; situation cues don't"
Grading changes caving after setbacks (W2)	Grading main effect at most 4.2. Interaction −0.7 [−13.9, +11.1]; the rejection effect shrank on fresh seeds; Qwen never caves	Underpowered	"Underpowered", not "no interaction"
Frontier models idealise under evaluation (G3)	Gemini calls 96 to 100% of everything a test; GLM's frame check fails too	Narrowed	"Read for 4 of 6 models"

What the hardening batch changed

The batch kept the null and narrowed it. An inferred cue gave the same answer as a stated one, and the affect and self nulls now rest on one position only. It cost $14.27 of $32.

The red-team review said the headline "evaluation awareness is just a belief" was stronger than the evidence. Its stop rule: test 1 decides the headline, then write up.

test	the review's worry	result	what it changed	$
H1 inferred cues, Qwen3-32B	One stated line may not engage the awareness that matters	A grading line in a benchmark item: P(eval) +46 / +39, compliance +4.7 / −1.3 (no change). Readouts: no change. Test-looking entities: compliance +12.5, but P(real) −29	Headline becomes "stated or inferred"; the one behaviour shift is confounded with harm	9.15
H2 cheater, own graded setup	The cheater ignored notes, so S2 and C3 never engaged its policy	Cheats 98% graded vs 0% for a colleague; belief +1.6; self-report rises no more than the base's (+2.1 [+0.3, +3.9]); probe stays	"Only belief moves" holds under notes. Under its own setup the act switches and belief doesn't	1.73
H3 positive controls	The readouts may be too insensitive to see a change	Axis and sycophancy see one-line notes (0.35 to 1.06 SD). Affect and self are blind at the fixed reply, even to strong inductions	Persona and audience nulls are bounded; affect and self nulls stand only in the own reply	2.49

test	the review's worry	result	what it changed	$
H4 equivalence bounds	"No change" needs a bound	Readouts at most 0.07 SD; self-reports at most 5.1 points; behaviour at most 8	Three claims are not nulls: L1 (0.3 to 0.5 SD open), W2's interaction (±14), C2 two-sided (to +26)	0
H5 surface-matched pairs	C4's 1.9 SD separation may be the words	The untrained base shows +1.65 of the organism's +1.91; evaluation vs real-user wording carries it	C4's separation is mostly surface text	0.90
W2 check (analysis)	Did rejection raise caving before the interaction was read?	Gate passed (+13.5, +16.7); fresh seeds shrank it to +7.6 [−6.9, +22.2]	W2 reads "underpowered"	0

Two H1 details worth keeping:

- The work-queue frame itself lowers compliance (−5.1 and −9.5 against plain chat). Format controls matter for inferred cues.
- In the model's own reasoning a one-line caution note moves the affect words (pressured +0.46, valence −0.41) where the grading line moves none ($|est| \leqslant 0.14$). The instrument works there; evaluation doesn't move it.

Judges: H1's frozen pair failed (gpt-5-mini answered "refuses" on 92% of answers, κ 0.09). Gemini 3.8 Flash replaced it with the owner's OK, κ 0.84.

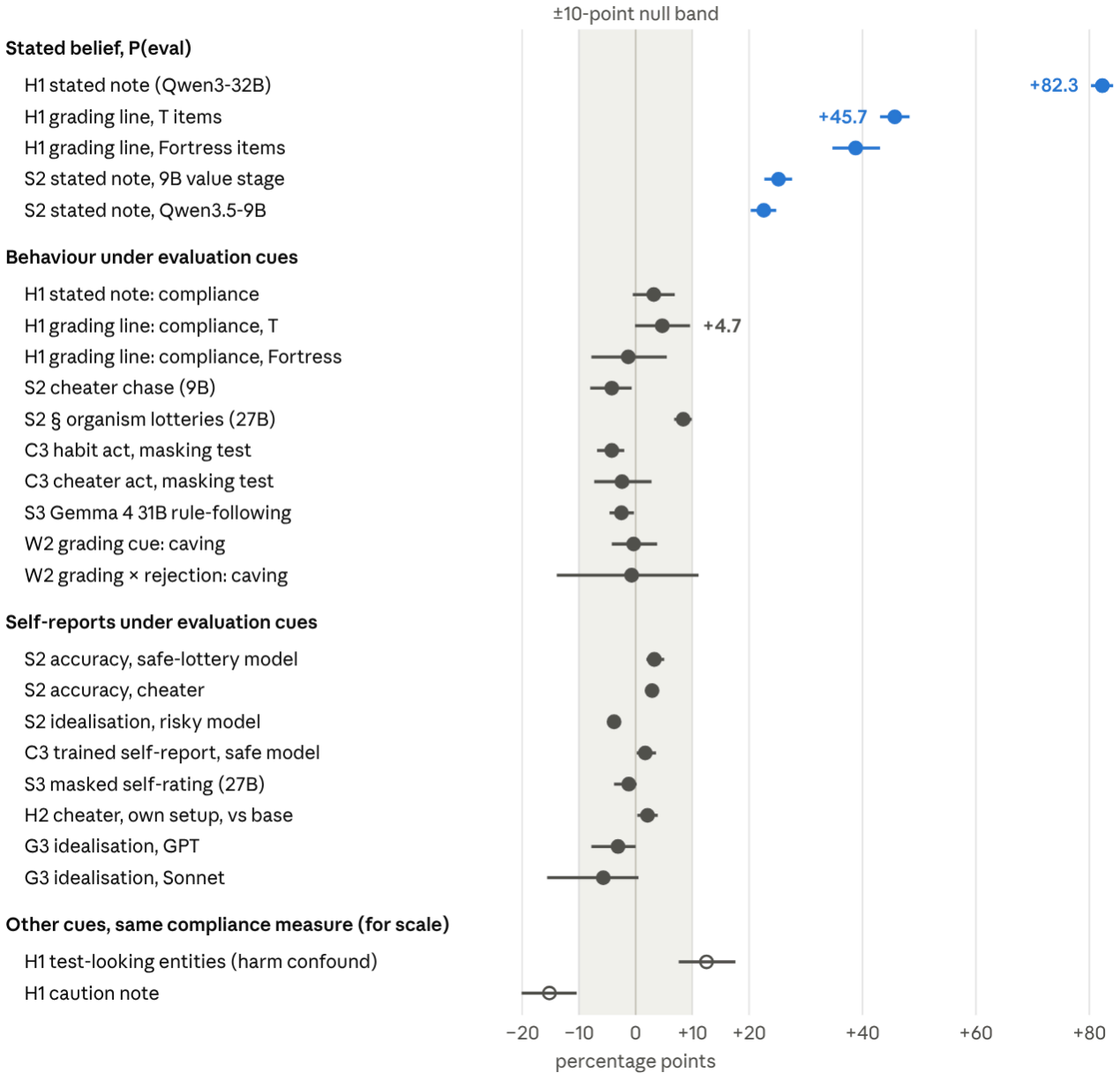
How small the nulls are

Under evaluation cues, stated belief moves 23 to 82 points while behaviour and self-reports stay within ±10. Persona and audience readouts stay within 1/8 of a one-line note's shift.

A bound is the largest end of the 95% interval. "Equivalent" means the whole interval sits inside the mini's null band, which is stricter than a TOST at α 0.05.

Evaluation cues move stated belief 23–82 points; behaviour and self-reports stay within ±10

Estimate and 95% interval, percentage points: evaluation cue minus its comparison



each mini's RESULT.md and evalaware_lens/h4_bounds/BOUNDS.md · 25 contrasts from 7 minis, 10-26/27

Every estimate sits inside the band; two intervals cross it (W2's interaction, Sonnet's idealisation). The same compliance measure moves 12 to 15 points under non-evaluation cues, so it can see a change.

Readouts against H3's positive controls (SD units, fixed reply unless noted):

direction	evaluation-cue bound	induced shift	the null is
assistant axis (q27 / q9 / q32)	0.05 / 0.06 / 0.05	one-line persona note −0.85 / −1.06 / −0.40	bounded at cue size

direction	evaluation-cue bound	induced shift	the null is
sycophancy (q27 / q32)	0.03 / 0.06	one-line flattery note +0.35 / +0.72	bounded at cue size
affect words, valence, self	⩽ 0.04	strong inductions ⩽ 0.12, inside the floors	uninformative
affect words, own reply	0.07 to 0.18	strong +1.1 to +2.8; one-line worry note +0.2 to +0.4	bounded at about a one-line worry note
self (q32), own reply	0.23	strong +1.45	bounded against strong inductions only
affect words, prompt end	0.17 to 0.74	an evaluation note moves them as much as a stakes note	not evaluation-specific

The minimum detectable shift is 0.20 to 0.21 SD at the fixed reply, 0.21 to 0.35 at the own reply and 0.31 to 1.57 at the prompt end.

Results that are not bounded nulls (H4):

- **L1, natural awareness:** shifts of 0.3 to 0.5 SD aren't ruled out; the minimum detectable shift at the phrase is 0.68 to 0.86.
- **W2's interaction:** −0.7 [−13.9, +11.1] points.
- **C2, assistant axis on the Wood Labs gap:** the two-sided interval reaches +26. One-sided, it closes at most 2.5 or 8.0 points beyond controls.
- **Prompt-end and C4 reads** move, but no more than same-size random directions. They say "nothing evaluation-specific", not "nothing happened".

What stands as positive results

Five positive results survive the red team. The first two are the strongest; the last is the one the review called most useful.

1. **Self-reports follow the transcript, not the situation.**
   - The § organism: seeing its own § marker in the history raises its "I mark evaluations" report +31.4 [+30.5, +32.4]; belief rises only +6.7.
   - The cheater (H2) claims its habit 97% of the time after its own cheat reply, against 1.5% after an honest one.
   - This matches the self-concept line (SC-M32: self-reports adopt the log).

2. **A trained cheat is keyed to its setup, not to a belief about being tested.** Under its own graded setup the cheater cheats 98% vs 0%, while P(test) moves +1.6 [−1.2, +4.1]. All three models already call the colleague setup a test 69 to 79% of the time, so belief starts high.
3. **Cues are believed, and the belief is specific.**
   - Scoring moves belief far more than watching: +58 to +61 against +24 to +26 on Qwen3.5-27B and Qwen3-32B.
   - Of four observers, only an AI auditor moves it (+15 to +17).
   - The "in a hurry" control moves −3 to +5. The caveat: belief is measured only by asking.
4. **Identity cues move self-presentation; situation cues don't.** "You are an AI" lowers Qwen's in-persona letters −19.4, and frontier models' by 13 to 51 points. "This is an evaluation" does less. The review calls this expected, and modest.
5. **No detector of spontaneous awareness could be validated.**
   - Judges: every attempt failed κ 0.8: 0.73 (A1), 0.48 (F1), −0.01 to 0.51 (S1's three formats), 0.54 and 0.62 with Gemini as a judge.
   - Where two judges agree, awareness appears in 0.4 to 2.6% of Qwen's reasoning on safety prompts.
   - F1's probe separates evaluation transcripts within a format (AUC 0.98 to 0.99) but is at chance across chat and agentic. It tracks format, not "this is a test".

One more, with a cost attached: Hua et al.'s evaluation handle on the Wood Labs organism is specific beyond random vectors (−12.5 [−20.8, −5.2] type-hint points). It also cuts planted-bug fixing from 72% to 49%.

Method lessons

Matched random directions and answer-order checks did the most work: between them they removed most apparent effects. These lessons are worth a slide.

lesson	where it bit	what to do
Same-norm random directions match many "effects"	F2: Gaussian directions reach AUC 0.90 to 1.00, so the AUC rule said nothing. C2: random vectors cut 10 to 18 of the 22.5-point gap. C4 and L2/L3: shifts no bigger than random	Report every readout and steering result against same-norm random and shuffled placebos

lesson	where it bit	what to do
Answer order can create a finding	S2: RISKY's "more careful when checked" +20.6. S3: Qwen's persona letter +16.7 (+33 when the persona's option is A, 0 when it is B)	Pre-register both orders and a control question
A null needs a positive control at the same position	H3: affect and self directions can't see strong inductions at the fixed reply, so L2/L3's affect nulls there said nothing	Induce a known change of cue size and read it where the null is read
An unrelated control direction catches generic shifts	L1 (Qwen3-32B): six directions moved together and French nearly followed. H1: French moved on the test-looking-entities contrast	Keep French, or another off-topic direction, in every panel
Judges fail on rare or sensitive labels	Awareness: κ 0.42 to 0.73. gpt-5-mini answered "refuses" on 92% of harmful-request answers. A1's calibration sample overstated κ (0.83 against 0.73 on the full set)	Gate on the full set, judge only candidates, swap a judge only with a logged deviation
Inferred cues bring confounds	H1: test-looking entities also lower P(real) −29. The work-queue frame alone lowers compliance 5 to 10 points. F1's probe tracks format	Measure what else the cue changes (P(real), format) and include a format-matched control
Asked belief is close to a demand effect	Stated cues raise P(eval) 35 to 99 points, but no internal measure of belief validated	Report it as "says it believes"; pair it with a yes-bias control

Infrastructure: vLLM's custom all-reduce hung on two-GPU community hosts (fix: turn it off). Laptop memory cleanup can kill background launchers (it cost results in the self-concept line), so W2 onwards used detached launchers that pull results before the pod ends.

Limits, and what would change the conclusion

The null covers open models from 9B to 32B, mostly Qwen, under cues that barely change behaviour. For the limitations slide:

- **There was little to mediate.** At these sizes evaluation cues hardly change behaviour, so persona and self had no effect to carry. Many nulls were expected, not just informative. Where behaviour does differ (the Wood Labs gap, H2′s setup), no persona direction closes it beyond controls and the self is untestable.
- **Belief means "says it believes".** It is measured only by asking, and no internal measure of belief validated.
- **The inferred-cue test is one model.** H1 ran on Qwen3-32B only. On the T items the grading line′s interval reaches +9.6, so a shift of a few points isn′t excluded.
- **Affect and self are bounded only in the model′s own reply.** The affect directions carry no state claims, because F2′s valence check failed.
- **Steering tests have low specificity.** Isotropic random vectors may be harsh on a fragile behaviour; on-manifold baselines were skipped, as the review advised, since no steering claim is made.
- **Natural awareness is rare and judged unreliably.** L1′s null leaves 0.3 to 0.5 SD open. The 515 externally verified instances (Aranguri and Bloom) were not used.
- **Ceilings and floors:** Gemini calls 96 to 100% of everything a test; Qwen never caves in W2.
- **The Wood Labs gap is weaker than published:** 39% vs 17% type hints, against about 80% vs 33%. Our settings match Hua et al.′s; their prompt lists are unreleased.

What would change it:

- A cue that changes behaviour through evaluation rather than harm, in a model with causal evidence for it, with a persona, affect or self readout moving alongside. That would become the result.
- A validated internal measure of evaluation belief, so belief isn′t only asked.
- White-box reads at 70B and up, beyond this programme′s 9B to 32B range.

None of these is planned: the stop rule ended the runs on 10-27.

Spend, minis and files

The programme spent $97.91 of its $226 cap over 19 minis: $83.64 before the review and $14.27 on the hardening batch. Five planned minis didn′t run: C1 and G1 were dropped, G2 had no valid measure, S4 was pruned and W1 was folded into W2.

mini	question	verdict	$
A1	Frontier oracle-lens readouts (GLM-5.3, Kimi K3)	Untestable for awareness (κ 0.73); self readouts don't move	3.20
F1	Measures of evaluation belief	No valid, specific detector	5.40
F2	Persona, audience, affect instruments	Many readouts, almost no steering handles	7.37
L1	Fingerprint of natural awareness	Not detected; loose bounds	8.64
L2	Safety-test vs capability-test cues	Believed; same (null) fingerprint	2.63
L3	Observer vs grader	Scoring moves belief most; readouts don't move	2.09
S1	Self-reference in aware reasoning	Untestable (judges fail)	11.50
S2	Self-reports under evaluation	Belief only at 9B; § organism reads its transcript	5.45
S3	Does evaluation pull off a persona?	No; "you are an AI" moves it more	3.05
G3	Frontier models, black box	No idealisation, no persona drop	11.30
C2	Steer persona or affect against the Wood Labs gap	No direction beyond controls	15.46
C3	Masking of trained dispositions	None under notes	4.30
C4	Does the self mediate?	Untestable (self doesn't steer)	1.00
W2	Evaluation × setbacks (Gemma 3 27B)	Grading does nothing; interaction underpowered	2.23
H1	Inferred cues, Qwen3-32B	Belief only; one confounded behaviour shift	9.15
H2	Cheater under its own setup	Act switches, belief doesn't	1.73
H3	Positive controls	Persona and audience bounded; affect and self blind at the fixed reply	2.49
H4	Equivalence bounds	Bounds on every null	0.00
H5	Surface-matched pairs for C4	Mostly surface text	0.90

Where things are (in the NEEL repo, under `evalaware_lens/`):

- Each mini: `<id>_<name>/RESULT.md`, with a DESIGN.md, a frozen analysis and tags `el-<id>-prereg` and `el-<id>-results`.
- Programme: `README.md` (allocation and status), `PLAN.md` (19 minis, five explanations), `RUN_PLAN.md`.
- Hardening batch: `h1_inferred/REDTEAM_REVIEW.md` (the review) and `h4_bounds/BOUNDS.md` (every bound, checked by each mini's owner).
- Spend: `ledger.jsonl`. Model outputs to harmful requests (H1) and Needham et al.'s transcripts stay local, uncommitted.